Subject: IT 241 Digital Electronics Date:24/06/2019

Sr. No	**Aim**
1	Getting familiar with various digital integrated circuits of different logic families. Study of data sheet of these circuits and see how to test these circuits using Digital IC Tester.
2	Digital IC Testers and Logic State Analyzer as well as digital pattern generators should be demonstrated to the students.
3	Configure diodes and transistor as logic gates and Digital ICs for verification of truth table of logic gates.
4	Configuring NAND and NOR logic gates as universal gates.
5	Implementation of Boolean Logic Functions using logic gates and combinational circuits. Measure digital logic gate specifications such as propagation delay, noise margin, fan in and fan out.
6	Study and configure of various digital circuits such as adder, subtractor, decoder, encoder, code converters.
7	Study and configurations of multiplexer and demultiplexer circuits.
8	Study and configure of flip-flop, registers and counters using digital ICs. Design digital system using these circuits.
9	Perform an experiment which demonstrates function of 4 bit or 8-bit ALU.
10	Introduction to HDL. Use of HDL in simulation of digital circuits studied in previous sessions using integrated circuits. Illustrative examples using FPGA or CPLD boards.